

Inventory Management System

An **Inventory Management System** developed as a learning and practical project using **Python and Streamlit**.

The goal of this project is to create a simple system for managing inventory, products, stock, and related information.

Project Status

Current Status: In Development

This project will be improved step by step as new features are added.

Technologies Used

- Python
 - Streamlit
 - Pandas
 - NumPy
 - Matplotlib
 - Git
 - GitHub
 - VS Code
-

Project Structure

```
inventory-managemnet-  
├── README.md  
├── streamlit/  
│   ├── 01_app_test/  
│   │   ├── app.py  
│   │   └── workflow.drawio  
└── ...
```

Installation

1. Clone the Repository

```
git clone https://github.com/aneelaaltaf007/inventory-managemnet-.git
```

2. Go to the Project Folder

```
cd inventory-managemnet-
```

3. Create a Virtual Environment

Using Conda

```
conda create -n inventory python=3.12
```

Activate it:

```
conda activate inventory
```

Or using Python venv

```
python -m venv inventory_env
```

Activate on Windows:

```
inventory_env\Scripts\activate
```



Install Dependencies

If you have a `requirements.txt` file:

```
pip install -r requirements.txt
```

If you don't have one yet, you can install the main packages:

```
pip install streamlit pandas numpy matplotlib
```

Run the Streamlit Application

Go to the app folder:

```
cd streamlit/01_app_test
```

Run:

```
streamlit run app.py
```

Or:

```
python -m streamlit run app.py
```

The application will open in your browser.

Git & GitHub Commands

These commands will be useful whenever you continue working on the project.

Check Git Status

```
git status
```

See Current Branch

```
git branch
```

See Remote Repository

```
git remote -v
```

Download Latest Changes

```
git pull origin main
```

Add All Changes

```
git add .
```

Commit Changes

```
git commit -m "Your commit message"
```

Push Changes to GitHub

```
git push origin main
```

Normal Git Workflow

Whenever you make changes to your project:

```
git status
git add .
git commit -m "Describe your changes"
git push origin main
```

For example:

```
git add .
git commit -m "Add inventory dashboard"
git push origin main
```

Branch Commands

Create a new branch:

```
git checkout -b feature-name
```

Switch to another branch:

```
git checkout main
```

See all branches:

```
git branch
```

Push a new branch:

```
git push -u origin feature-name
```



BACK Undo / Recovery Commands

See previous commits:

```
git log --oneline
```

See changes that have not been committed:

```
git diff
```

Undo changes in a file that have not been committed:

```
git restore filename
```

Unstage a file:

```
git restore --staged filename
```



Useful Git Commands

Check the current repository:

```
git status
```

Check commit history:

```
git log --oneline
```

Check remote URL:

```
git remote -v
```

Update your local repository:

```
git pull origin main
```

Push your work:

```
git push origin main
```



Planned Features

Future versions may include:

- ☐ Add products
- ☐ Delete products
- ☐ Update products
- ☐ Search products
- ☐ Stock management
- ☐ Low-stock alerts
- ☐ Sales management
- ☐ Purchase management
- ☐ Dashboard
- ☐ Data visualization

- ☐ Database integration
- ☐ User authentication
- ☐ Admin panel
- ☐ AI-based inventory predictions
- ☐ NLP-based inventory assistant

Future AI Features

As an AI student, this project can later be expanded with AI features such as:

Machine Learning

- Demand prediction
- Sales forecasting
- Stock prediction
- Product recommendation
- Inventory optimization

NLP

An NLP assistant could allow users to ask questions such as:

"Which products are low in stock?"

"Show me products sold this month."

"Which product has the highest sales?"

Future Database

The project can later use a database such as:

- MySQL
- PostgreSQL
- SQLite

Possible tables:

Products
Customers
Suppliers
Sales
Purchases
Users
Inventory

Testing

Before pushing changes to GitHub:

```
python -m py_compile app.py
```

Check whether Streamlit starts correctly:

```
python -m streamlit run app.py
```

Requirements File

After installing your project packages, you can create:

```
pip freeze > requirements.txt
```

Then another person can install the same dependencies using:

```
pip install -r requirements.txt
```

Saving Your Work

Before closing your project, check:

```
git status
```

Then:

```
git add .  
git commit -m "Save project progress"  
git push origin main
```



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License

This project is created for educational and learning purposes.



Project Goal

The long-term goal is to develop this project into a complete **AI-powered Inventory Management System** that combines:

Python + Streamlit + Database + Machine Learning + NLP

```
Inventory Management  
  ↓  
  Database  
  ↓  
  Data Analysis  
  ↓  
  Machine Learning  
  ↓  
  AI Prediction  
  ↓  
  NLP Assistant
```



Important Note

Keep this README updated whenever you add a major feature to the project.

For example:

```
git add .  
git commit -m "Add product search feature"  
git push origin main
```